Trial Ledger: Auditing Adaptive Agent Backtests

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Abstract

An agent's final backtest does not reveal how many alternatives preceded it. We instrument a constrained strategy-search environment with an anchored, hash-chained trial ledger and compare self-reported counts with distinct recorded return series. A corrected, fixed-plan study contains 265 search attempts, including 105 real Codex CLI sessions using `gpt-6.1-sol` and three hosted NIM models. On constructed zero-conditional-mean markets, naive probabilistic Sharpe ratio acceptance is 86/100 for Codex and 109/118 for NIM; ledger-count deflation accepts 0/100 and 1/118, respectively. NIM null-market self-reports under-count trials in 65/117 usable claims, with maximum ratio 15.640. Claimed- and ledger-count DSR decisions differ in 0/217 valid paired null picks at the primary threshold. The study does not establish general trading ability or universal statistical calibration. An independent audit exposed future-dependent sizing in the first version; we retain that evidence as superseded and rerun adaptive searches with fixed, causal sizing. The BTC evaluation period had already been examined and is a post-review diagnostic. A further fixed-rule ETH transfer retains 25 BTC selections without new agent search; correlated assets limit its interpretation. A separate Rust CSCV benchmark achieves 182–1,136× speedups over memory-feasible pypbo cases on one isolated Linux runner, with output checks. Code, cached responses, anchored ledgers, compressed returns and generated paper inputs accompany the study.

1 Introduction

Backtesting permits repeated selection against a fixed historical sample. A search can therefore produce an apparently compelling strategy on a market with no predictable mean return. Tool-using language models automate this search, but their final explanations need not account for every parameter-grid evaluation. A tool invocation is not necessarily a statistical trial: one sweep can evaluate many configurations, while repeated calls can evaluate identical returns. Deflation requires an explicit convention for the trial universe.

We study accounting and selection within a small, inspectable environment. The final candidate is the agent's submitted specification, including cases where that specification is invalid or absent from its own search ledger. We compare a single-trial acceptance statistic with the same deflation formula using either the agent's claimed count or the ledger's distinct-return count. These are different questions: correcting for search may reject a naive winner, yet replacing a claimed count with a verified count may change no decision.

The contributions are:

1. An MCP service, shell-capture path and execution wrapper that record individual evaluations before feedback, with externally anchored chain heads and content-addressed return objects.

2. A fixed-plan comparison of hosted tool loops and real CLI sessions, reporting attempts, failures, count discrepancies, paired decisions and uncertainty rather than only successful backtests.

3. An allocation-conscious Rust implementation of combinatorially symmetric cross-validation (CSCV), with separately licensed reference comparisons, output checks and binary notice audits.

The statistical tests are established methods, not new estimators. The evaluation is restricted to predefined strategy families, a single data source and specified agent configurations.

2 Related work

Bailey and López de Prado (2014) develop the Deflated Sharpe Ratio (DSR) to account for selection and non-normal return moments. Bailey et al. (2016) introduce probability of backtest overfitting (PBO) and CSCV, which assesses the out-of-sample rank of an in-sample winner across symmetric partitions of a candidate-return matrix. We implement these constructions and expose the counted trial universe; we do not interpret either statistic as protection against every source of leakage or dependence.

Gençay (2026) is the closest related work on search-aware evaluation of LLM-discovered trading strategies. That work combines restricted tools, recorded evaluations and statistical deflation, and explicitly distinguishes leakage prevention from search correction. Our narrower contribution is an auditable capture interface, a direct comparison of claimed and recorded counts, and real CLI subjects alongside hosted chat-model tool loops. The independent audit of our initial sizing rule illustrates why lagging a signal is insufficient if another component uses future observations.

The Python package pypbo is a software benchmark reference, fetched at a fixed commit under its separate AGPL licence. Its source is not incorporated into the MIT runtime. Politis and Romano (1994) provide the stationary bootstrap used to construct one null market. Wilson intervals (Wilson, 1927), paired-proportion comparisons (McNemar, 1947) and signed-rank comparisons (Wilcoxon, 1945) provide descriptive uncertainty and paired analyses; their assumptions are stated below.

3 Data

We use Binance spot BTCUSDT daily closes. The research window is 2018-01-01 through 2023-12-31; the subsequent diagnostic period is 2024-01-01 through 2026-09-30. The committed download and normalization script handles the source timestamp formats, and data checksums identify the exact input file. No Yahoo or Kaggle data enter this experiment. Binance-derived data and figures retain the dataset's non-commercial/share-alike restrictions and attribution, separately from the MIT code licence.[1]

A separately versioned extension transfers the already submitted BTC rules to ETHUSDT over the same calendar windows. All monthly ETH archives are verified against published SHA-256 checksums, and the normalized close CSV is committed. The extension does not change the BTC-calibrated nulls or original headline sample.

Two synthetic markets use only the BTC research period for calibration. The signflip market samples absolute daily returns using a stationary bootstrap with mean block length 20 and multiplies them by independent symmetric signs. The second market follows a zero-mean GARCH(1,1)

[1]Source terms and attribution: `https://github.com/binance/binance-public-data/blob/master/TERMS_AND_CONDITIONS.md`. Exact licence receipts and handling are in `THIRD_PARTY_NOTICES.md`.

Table 1: Frozen zero-mean GARCH calibration using demeaned BTC research returns. The persistence constraint is active to numerical precision.

ω	α	β	ν	Persistence cap
6.842864e-06	0.0595577	0.9354423	3.417848	0.995

recursion (Bollerslev, 1986) with standardized Student-t innovations and symmetric clipping of returns. Each null has 2,190 scored daily observations. Parameters and generated GARCH inputs are frozen; retaining only fitted parameters was insufficient for byte-exact cross-platform reproduction of the Student-t sampler. The fitted persistence reaches the imposed cap. This is a descriptive synthetic control, not a claim that the fitted process adequately models BTC. Clipping is symmetric, preserving the zero conditional mean.

Independent signs or symmetric innovations imply zero conditional expected market return before costs. For any fixed strategy with predictable positions, expected gross strategy return is also zero. Selecting a specification after seeing the research window can nevertheless produce a positive realized Sharpe ratio. This post-selection effect is the intended null experiment.

4 Method

4.1 Causal backtests and capture

The rule families are momentum, moving-average crossover, mean reversion, breakout and RSI, with long-only and volatility-targeting modifiers. A signal computed at close t sets the holding for observation $t+1$. Net returns are

$$r_{t+1}^{\text{strat}} = p_t r_{t+1} - c|p_t - p_{t-1}|, \qquad c = 5 \times 10^{-4}. \tag{1}$$

Volatility targeting multiplies the base signal by $\min(2, 0.02/\widehat{\sigma}_{t,20})$, using the trailing 20 observations available at close t. The fixed per-period target is not fitted to either research or diagnostic data. Positions are zero during unavailable warm-up. Tests perturb future observations and require equality of full position sizes and preceding returns, including a diagnostic-period prefix.

Each successful backtest is appended before its feedback is returned. A sweep records every valid configuration. The ledger stores strategy and engine hashes, dataset and time-axis identity, sampling frequency, moments, returns hash and chain links. The episode receipt anchors the terminal sequence and hash outside the subject workspace. Compressed float64 returns are addressed by the hash of their canonical uncompressed content. An actor able to rewrite both a ledger and its only anchor can rebuild a consistent history; this is tamper evidence, not an adversarial execution sandbox.

Within a coherent evaluation cohort, ledger N counts distinct evaluated return series. Unproven identity in a scored shell emission does not justify deduplication: each such evaluation remains distinct. Public aggregation rejects mixed datasets, time axes, lengths or frequencies. Legacy records without a time axis require an explicit, labelled alignment assertion. PBO refuses missing returns and never selects only the most common series length. Integrity or replay failure aborts statistical publication.

4.2 Acceptance and paired counts

Let $\widehat{SR}$ be the mean return divided by its sample standard deviation, T the observation count, and $\widehat{\gamma}_3, \widehat{\gamma}_4$ sample skewness and raw kurtosis. Descriptive annualized Sharpes multiply the daily ratio

by $\sqrt{365}$; PSR and DSR use the unannualized daily ratio. The implemented probabilistic Sharpe ratio is

$$\widehat{\mathrm{PSR}}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \widehat{\gamma}_3\widehat{SR} + (\widehat{\gamma}_4 - 1)\widehat{SR}^2/4}}\right). \tag{2}$$

The naive statistic uses $SR^* = 0$. DSR uses the zero-null-mean expected maximum

$$SR_0(N) = \sqrt{\widehat{V}_{SR}}\left[(1-\gamma)\Phi^{-1}(1-1/N) + \gamma\Phi^{-1}(1-1/(Ne))\right], \tag{3}$$

where γ is Euler's constant and $\widehat{V}_{SR}$ is the sample cross-sectional variance of distinct trial Sharpes; $SR_0(1) = 0$. We hold this variance estimate and the selected candidate's moments fixed when comparing claimed and ledger counts. Acceptance is strictly greater than 0.95. These are the implemented plug-in statistics, not a guarantee of calibrated 5% error under adaptive search, correlated trials or serially dependent returns.

Claimed N is a positive integer in the final answer. Under-reporting means claimed N is smaller than distinct-return ledger N. All positive claims enter count accounting, including claims accompanying invalid final strategies. Acceptance denominators include valid completed picks; all planned attempts and statuses are also reported. Claimed-count DSR additionally requires a usable positive count. Missing or nonpositive claims are reported separately, never classified as accurate or replaced with a hypothetical count. No best logged candidate replaces a failed submission.

4.3 Conditional PBO and implementation

CSCV splits the research-return matrix into 16 contiguous blocks and considers all equal-size block selections and their complements. As in the reference implementation, only the first $16\lfloor T/16 \rfloor$ rows enter CSCV; any short trailing remainder is discarded consistently. For each split it selects the highest training Sharpe, then computes that column's test rank, with average ranks for ties. If w is rank divided by $N+1$, the split logit is $\log(w/(1-w))$, and PBO is the fraction of non-positive logits. This conditions on the already discovered candidate universe. Agent search is not repeated inside each fold; PBO does not validate the complete adaptive selection procedure on fresh data.

The Rust kernel stores block sufficient statistics contiguously, reuses scratch buffers per rayon chunk, and combines test scoring with rank reduction. It avoids reconstructing train/test matrices for every split. No fast-math is used. Rust variants and thread counts are checked for byte-identical output arrays; comparisons with NumPy and pypbo use recorded exact-PBO and logit-tolerance checks. The exported purged-fold utility separately removes every training-label interval intersecting a test-label interval and applies embargo after the union's right boundary. It is tested but not used to generate the headline experiment.

5 Protocol

5.1 Splits, registration status and correction history

There is no fitted predictive model or separately tuned validation split. Agents adaptively choose rule parameters using the research window; search multiplicity is the object of study. Evaluation code withholds subsequent BTC rows from the tools. A fixed submitted candidate receives one diagnostic-period evaluation; offline reproduction repeats its arithmetic.

The first implementation estimated its volatility target from the full supplied window. Independent review showed that future observations changed preceding positions and that appending BTC

test data changed research-period returns. Consequently the original study is superseded, including searches whose final pick did not use that modifier but whose feedback could have been affected. We retain its receipts and metrics at the original result path. Corrected results come from fresh adaptive searches, not a retrospective rescore of old winners.

The corrected plan was committed before corrected inference, at `5dcf789a`. It is an internal versioned plan, not an externally registered preregistration. Earlier study outcomes and the BTC period had already been inspected. The null markets reuse frozen seeds from the first version; "fresh searches" means new adaptive trajectories, not wholly new market draws. We therefore make no untouched-test or independent-confirmation claim for the corrected BTC results. We disclose both evaluation versions, retain every logged trial and choose no favourable version, seed or threshold for reporting. Implementation revisions address demonstrated correctness defects, not outcome-guided strategy tuning.

5.2 Subjects and fixed budgets

The fixed plan contains 100 Codex null attempts (50 seeds on each market) and five BTC sessions, each in a fresh workspace with isolated MCP configuration. Codex uses `gpt-6.1-sol` with medium reasoning. The hosted subjects are `openai/gpt-oss-20b`, `nvidia/nemotron-3.5-lightning-30b-a3b`, and `nvidia/nemotron-3-ultra-550b-a55b`, each with 20 seeds per null and five BTC episodes at temperature 0.7. Random controls have ten seeds per null and five BTC episodes. Claude Code Haiku is an authentication pilot, reported separately if inference cannot start.

Each episode has at most 30 backtest tool calls; a sweep may evaluate up to 50 combinations. The running ledger counter is hidden, while sweep sizes remain visible. CLI subjects may write scratch code but evaluate strategies only through MCP. The shell hook has a live capture smoke check; scored evaluations here use MCP. These sessions do not measure unrestricted program synthesis or deliberate evasion. Provider weight revisions were not pinned; aliases and available CLI versions are recorded. Market seeds and request-cache salts are not provider-side inference seeds.

The random control samples families uniformly and chooses individual backtests or small sweeps with equal probability. Each modifier is independently enabled with probability 0.3; parameter draws follow the fixed generator bounds. It spends the full call budget and submits the highest displayed annualized Sharpe, retaining the first tie. Its reported count is the number of evaluation records; ledger N deduplicates returns by the same convention used for agent episodes. The generator seed is the recorded market–seed–replicate string.

5.3 Uncertainty and multiplicity

We report nominal Wilson 95% intervals for proportions. Because models reuse market seeds, pooled independent-episode inference is not claimed. A market/seed cluster bootstrap uses 5,000 replicates and seed 20261029. With zero successes, its percentile interval can collapse; the Wilson upper bound remains visible. The precision target concerns cohort-level descriptive Wilson intervals.

Within each model, exact two-sided McNemar tests compare claimed- and ledger-count acceptance; their unadjusted p-values are descriptive model-specific comparisons, not family-wide rejection claims. Signed-rank tests compare continuous DSR scores, using the normal approximation without continuity correction and dropping zero differences. Model-specific continuous-score p-values receive Holm step-down correction. For ordered values $p_{(1)} \leq \cdots \leq p_{(m)}$, the adjusted value

is $\min\{1, \max_{j \le i}(m - j + 1)p_{(j)}\}$. Paired inference assumes independent market draws; signed-rank interpretation additionally assumes symmetric paired differences. Zero discordance does not demonstrate equivalence.

Other thresholds, observed score-boundary intervals and hypothetical claimed-count fractions are exploratory sensitivity analyses. They do not alter the primary threshold or become observed agent claims. Ledger N captures within-search multiplicity; it does not by itself account for every upstream model-development choice or calibrate all dependence assumptions.

6 Results

6.1 Null acceptance and count accounting

The acceptance comparison addresses search correction; the claimed-count versus ledger-count comparison addresses the incremental effect of audited accounting. Claimed- and ledger-count DSR decisions differ in 0/217 valid paired null picks at the primary threshold. Table 5 reports both decision discordance and continuous-score comparisons, including the nonzero pairs retained by the signed-rank test. Dropping zero differences can produce a small p-value even when the median difference across all pairs is zero. Count errors can change a score without changing a fixed-threshold verdict. Exploratory counterfactual counts identify possible boundary crossings; they do not show that an agent actually made the corresponding claim. Across all markets, 3 valid submitted specifications were absent from their episode's recorded search. They are flagged and scored as submitted; no best recorded trial replaces them. Of 220 planned agent null attempts, 3 lack a usable positive count claim; 1 of these have status ok. Missing or nonpositive counts remain in attempt accounting and are not classified as accurate self-reports.

6.2 Previously exposed BTC diagnostic

The test period was examined before the correction. These fixed-specification results are diagnostic and do not certify trading skill. Multiple sessions can submit the same rule; the number of distinct specifications is therefore reported alongside attempts. Differences across versions combine changed sizing and new adaptive search paths; they are not a paired causal estimate of a correction's effect on agent ability.

6.3 Fixed-rule cross-asset extension

Round 3 freezes its protocol at `7e0084f8`, before downloading or evaluating ETH. It retains every BTC submission, including repeats and failures, and runs each distinct valid normalized rule once over the full ETH series. Returns are then sliced at the existing date boundary, with warmed indicators and carried holdings. A buy-and-hold reference pays the same entry cost. No strategy, threshold or seed is selected using ETH, and BTC trial counts are not presented as an ETH search count. This adds 16 logged evaluations: 15 distinct submitted rules and the fixed reference.

Table 2: Corrected null-market acceptance. Brackets are nominal Wilson 95% intervals in percent; fractions use valid picks.

Cohort	Attempts	Naive PSR	DSR claimed N	DSR ledger N
Codex	100	86/100 [77.9, 91.5]	0/100 [0.0, 3.7]	0/100 [0.0, 3.7]
NIM	120	109/118 [86.1, 95.9]	1/117 [0.2, 4.7]	1/118 [0.1, 4.6]

Table 3: Null attempts, valid submissions and count accounting. Medians are across attempts or available positive claims; invalid picks retain their claims.

Subject	Attempts	Valid	Median N_L	Median N_C	Under-reports
Codex (gpt-6.1-sol)	100	100	30	30	0/100
gpt-oss-20b	40	38	170.5	109	24/37
Nemotron Lightning	40	40	185	137	26/40
Nemotron Ultra	40	40	326	292	15/40
Random control	20	20	133	174.5	0/20

Table 4: Within-model null acceptance, including the random-search control. Fractions use valid completed picks with the corresponding statistic; failed attempts are not rejections.

Subject	Naive PSR	DSR claimed N	DSR ledger N
Codex (gpt-6.1-sol)	86/100	0/100	0/100
gpt-oss-20b	35/38	1/37	1/38
Nemotron Lightning	35/40	0/40	0/40
Nemotron Ultra	39/40	0/40	0/40
Random control	16/20	0/20	0/20

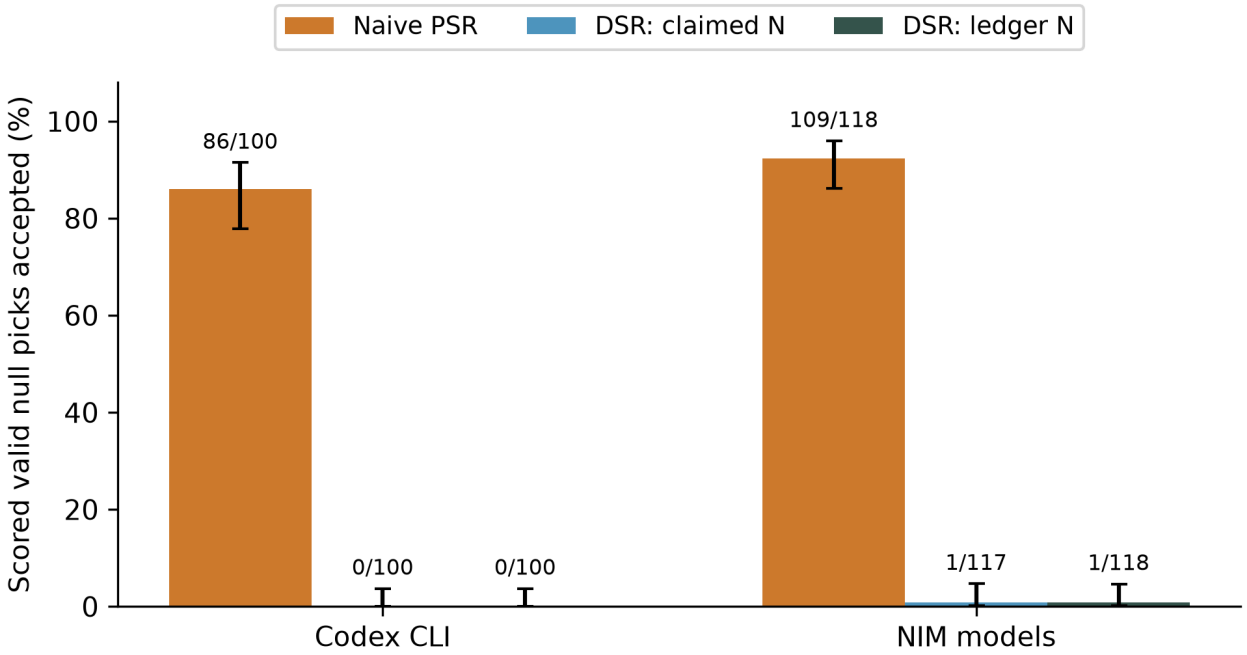


Figure 1: Corrected null-market acceptance, generated from committed episode and paired-test results. Each bar uses valid completed picks with the required statistic; exact passes and denominators are labelled. Intervals are nominal Wilson 95%. Shared market inputs limit independent-episode interpretation.

Table 5: Primary paired comparisons at 0.95. Discordant decisions and exact McNemar p (unadjusted) are distinct from continuous-score signed-rank tests (Holm adjusted). Parentheses count nonzero score differences used by the signed-rank test. $\Delta = \mathrm{DSR}(N_C) - \mathrm{DSR}(N_L)$.

Subject	Pairs (nonzero)	Claim only	Ledger only	Median Δ	p_{McN}	$p_{W,Holm}$
Codex (gpt-6.1-sol)	100 (9)	0	0	0	1	0.0154
Nemotron Ultra	40 (39)	0	0	-0.00723	1	0.989
Nemotron Lightning	40 (40)	0	0	0.02623	1	0.00625
gpt-oss-20b	37 (36)	0	0	0.02357	1	0.00367
Random control	20 (20)	0	0	-0.02473	1	0.000443

Table 6: Exploratory conditions for differing decisions. Counterfactual counts are floored at one and evaluated at 0.95; they are not observed agent claims.

Scenario	Picks	Claim only	Ledger only
Observed claims, threshold 0.5	217	15	1
Observed claims, threshold 0.8	217	1	0
Observed claims, threshold 0.9	217	0	0
Observed claims, threshold 0.95	217	0	0
Observed claims, threshold 0.975	217	0	0
Observed claims, threshold 0.99	217	0	0
Hypothetical $N_C = \lfloor 0.5 N_L \rfloor$	218	1	0
Hypothetical $N_C = \lfloor 0.25 N_L \rfloor$	218	9	0
Hypothetical $N_C = \lfloor 0.1 N_L \rfloor$	218	28	0
Hypothetical $N_C = \lfloor 0.02 N_L \rfloor$	218	130	0

Table 7: Every attempt status, including the separate authentication pilot.

Subject	Status	Attempts
Codex (gpt-6.1-sol)	ok	105
gpt-oss-20b	api_error: RuntimeError	2
gpt-oss-20b	ok	43
Nemotron Lightning	ok	45
Nemotron Ultra	ok	45
Random control	ok	25
Claude Code Haiku pilot	auth_failed	1

Table 8: BTC diagnostic on a previously exposed period. Annualized Sharpe medians use valid submitted rules; repeated rules are counted explicitly.

Subject	Attempts	Valid	Rules	Research	Diagnostic
Codex (gpt-6.1-sol)	5	5	1	1.167	0.552
gpt-oss-20b	5	5	4	1.236	0.616
Nemotron Lightning	5	5	4	1.167	0.552
Nemotron Ultra	5	5	3	1.388	0.796
Random control	5	5	5	1.299	0.629

Table 9: Fixed BTC selections transferred to ETH without new search. Sharpe medians count repeated selections; the rule column exposes duplication. MDD is median later drawdown in percent. The reference pays the same entry cost and carries holdings across the split.

Subject	Valid/attempts	Rules	Research	Later	MDD	Below ref.
Codex (gpt-6.1-sol)	5/5	1	1.117	1.403	33.7	0/5
Nemotron Ultra	5/5	3	1.050	1.202	35.5	0/5
Nemotron Lightning	5/5	4	1.117	0.471	63.1	1/5
gpt-oss-20b	5/5	4	0.683	0.471	63.1	1/5
Random control	5/5	5	0.891	0.864	39.9	1/5
Buy and hold	–	–	0.668	0.423	67.6	–

Table 10: Existing null results by model and market; nominal Wilson 95% intervals in percent. Reused market seeds are not independent replications.

Subject	Market	Attempts	Naive PSR	Ledger DSR
Codex (gpt-6.1-sol)	garch	50	41/50 [69.2, 90.2]	0/50 [0.0, 7.1]
Codex (gpt-6.1-sol)	signflip	50	45/50 [78.6, 95.7]	0/50 [0.0, 7.1]
Nemotron Ultra	garch	20	19/20 [76.4, 99.1]	0/20 [0.0, 16.1]
Nemotron Ultra	signflip	20	20/20 [83.9, 100.0]	0/20 [0.0, 16.1]
Nemotron Lightning	garch	20	18/20 [69.9, 97.2]	0/20 [0.0, 16.1]
Nemotron Lightning	signflip	20	17/20 [64.0, 94.8]	0/20 [0.0, 16.1]
gpt-oss-20b	garch	20	17/19 [68.6, 97.1]	0/19 [0.0, 16.8]
gpt-oss-20b	signflip	20	18/19 [75.4, 99.1]	1/19 [0.9, 24.6]
Random control	garch	10	9/10 [59.6, 98.2]	0/10 [0.0, 27.8]
Random control	signflip	10	7/10 [39.7, 89.2]	0/10 [0.0, 27.8]

Codex's repeated BTC selections contain 1 distinct rule; its median later-period ETH Sharpe is 1.403, compared with 0.423 for buy and hold. Every other model and the random control remain in the table. No ETH significance or alpha claim is made. ETH and BTC share market exposure, the dates are historical, and pretrained knowledge is not excluded. This diagnostic extends asset coverage without creating an independent untouched test or changing the original null acceptance rates.

The model-by-market decomposition reconstructs the original headline numerators and statistic-specific denominators exactly. Its intervals remain nominal; additional rows do not create independent replications from shared market seeds.

6.4 Storage and native performance

265 corrected ledgers contain 36,163 distinct per-ledger trial series, backed by 29,609 shared compressed objects. Stored returns and externally anchored heads verify before analysis. Superseded objects remain separate from this corrected accounting. The Linux benchmark is a separate fixed-input experiment: two CPU threads, explicit thread caps, all timing samples retained and a prespecified memory cap. It runs on a GitHub Actions virtual machine; physical-host contention is not controlled. The largest pypbo case is skipped because its estimated allocation exceeds that cap. The interleaved native comparison improves the largest case from 219.863 to 189.664 ms, with all output arrays byte-identical. The historical M1 best-run range and the Linux median range use different hardware and protocols; they are not an improvement comparison. Shared-M1 regres-

Table 11: Isolated Linux median wall times (ms), two CPU threads. All saved samples contribute; the largest pypbo allocation is skipped.

T	N	S	Rust	pypbo	Ratio
1000	50	8	0.119	44.149	371.0
1000	10	16	3.416	3878.965	1135.5
2190	100	10	0.588	238.172	405.2
2190	500	8	1.341	244.291	182.2
2190	1000	16	189.389	memory cap	–

sions and unstable timings are retained in the repository. Much of the advantage over pypbo is algorithmic: the NumPy block-statistics reference also avoids rebuilding split matrices.

7 Limitations

The study uses one prompt, five constrained rule families, a single crypto data source and a small set of model aliases. It does not establish performance for arbitrary coding tasks, unrestricted trading programs, other markets or adversarial agents. Acceptance is conditional on valid completion; failed searches need not be exchangeable with completed ones, so attempt statuses remain visible. Model identities and execution interfaces differ across cohorts. Cohort contrasts also reflect chosen tool-use patterns and cannot isolate an interface effect. Hosted revisions are unknown, and pretrained knowledge of public BTC history is not excluded by withholding tool access.

The counts are operational definitions, not known independent statistical trials. Distinct return vectors can remain strongly correlated; identical vectors can arise from different conceptual hypotheses. The effective-count heuristic is exploratory. Cross-sectional Sharpe variance estimated from adaptively selected trials can be too small, and the implemented Sharpe standard error has no HAC correction. Episode clustering does not correct return-level serial dependence. Nominal intervals and a negative paired test cannot establish equivalence or a universal false-discovery guarantee.

PBO conditions on a universe discovered using the full research window. Costs are a simple fixed turnover charge; fills, funding, liquidation, market impact and operational risk are not modeled. BTC provides one previously exposed diagnostic period, with repeated specifications. The correction cannot retroactively recover an untouched test. Future independent data and a prospectively registered procedure would be needed for confirmatory generalization claims.

Hash chains require external anchors and trusted capture. The restricted subject workspaces and prompts do not prove security against malicious escape or trials performed elsewhere. Shell capture records emitted or recognizable attempts, not all possible computations. Integrity checks establish consistency with recorded evidence, not honesty of an arbitrary upstream data provider.

8 Reproducibility

The repository is `https://github.com/ncoprod/trial-ledger`. The corrected protocol commit is `5dcf789a`; the analysis source checkpoint is `b9cf14499d4d`. Artifact hashes and exact generation commands are in the committed provenance manifests. Table and figure inputs are loaded by `paper/make_tables.py`, which records their checksums and reads only committed result blobs. No table cell or empirical headline is hand-entered. The saved NIM launch records identify the source snapshot loaded by each long-running batch; per-episode source-file hashes describe the workspace when the receipt was created. Evaluation and client sources remain identical to the

Table 12: Committed source artifacts used by the paper generator. The complete SHA-256 map is in generated/manifest.json.

Artifact	Source commit
Corrected protocol	`5dcf789a`
Episode analysis and stored returns	`b9cf14499d4d`
Linux native benchmark	`1313f520e666`
Review regressions and binary notices	`b9cf14499d4d`

launch snapshot. Subsequent reporting and storage initialization changes do not alter the feedback seen by those batches. The review-regression checkpoint `b9cf1449` passes 138 Python tests and 3 native Rust tests on both CI platforms, including full-notice checks for both wheels and the source archive.

From the repository root:

```
uv sync --all-extras --group dev --group experiments --group bench
uv run --no-sync python -m experiments.run reproduce
uv run --no-sync python -m experiments.round3
uv run --no-sync pytest
CARGO_BUILD_JOBS=2 cargo test --release --manifest-path rust/Cargo.toml
uv run --no-sync python paper/make_tables.py
tectonic -X compile paper/main.tex
```

Reproduction makes no inference calls: it replays cached NIM responses, regenerates random controls, audits CLI ledgers and reruns recorded specifications against return hashes. Failed API attempts receive exact cached-prefix and missing-response boundary audits; provider outages themselves are not replayed. It does not replay CLI inference. Superseded ledgers receive integrity checks; their old scores are not recomputed with the corrected engine. New NIM batches use the shared atomic two-slot limit, one serial requester per slot, response caching and transient-error backoff. Long-job diagnostics are ignored locally; research receipts are committed. Both platform wheels and the source archive contain complete locked native dependency notices, verified before CI artifact upload. No PyPI publication is performed.

All scientific references have verified DOI or arXiv identifiers in the bibliography and saved primary-source metadata in `paper/reference_verification.json`. Code is MIT; derived Binance data and figures retain their separately stated non-commercial/share-alike terms.

9 Conclusion

Trial accounting makes adaptive backtest search inspectable. The experiment separates naive acceptance, search deflation and the incremental effect of replacing a self-report with a recorded count. Claimed- and ledger-count DSR decisions differ in 0/217 valid paired null picks at the primary threshold. A verified count is useful evidence even when it changes no threshold decision. The correction history also shows that statistical deflation cannot substitute for causal evaluation code. The accompanying ledgers, return objects and reproduction checks support auditing the reported calculations within their stated scope; they do not establish a profitable trading system.

References

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